WEMAKEDEVS × AWS ENVIRONMENTAL HACKS · WASTE & ENERGY TRACK

Lane Scorecard
How the website works

Quick bin checks become specific feedback, a voice for cleaning staff, and a weekly check on whether anything actually improved.

Live site: **https://ss9ib2qbek.execute-api.us-east-1.amazonaws.com**
Guide version 2 (whole NSUT campus) · 9 October 2026 · All screenshots use the **Demo** workspace: every number shown is simulated.
Each screenshot has red letters (A, B, C…) with arrows. The table next to it explains each letter.

1. The problem it solves

Across NSUT, bins stand in pairs: green “Degradable waste (kitchen waste)” and blue “Non-degradable waste (recyclable)”. People mix them anyway.

- **Nobody knows how bad it is.** Bins are not checked regularly, so there is no record.
- **Nobody tells people what they did wrong.** Generic “please segregate” posters do not change behaviour.
- **Cleaning staff suffer.** They sort mixed waste by hand, sometimes with broken glass, blades or needles inside, and nobody asks them.
- **Good sorting can be wasted at pickup** if both bins are emptied into one cart, and nobody notices.
- **Nobody checks whether a fix worked.** Something is tried, then forgotten.

2. What the website does

1. Record	A volunteer or warden rates a green/blue bin pair in about two minutes.
2. Workers report	Cleaning staff use a picture-only Hindi page with voice: faces for each bin, pictures for what was wrong, a red button for dangerous items.
3. Score	Each location gets a weekly score out of 100. Bins nobody saw never count as clean.
4. Specific feedback	A shareable English/Hindi card: what went wrong and how often, why it matters, how other locations did, and one thing to try.
5. Safety first	Dangerous items flagged by staff become that location’s top action.
6. Did it work?	Next week says clearly better, clearly worse, or no clear change, and refuses to claim normal ups and downs as change.
7. Collection follow-up	If waste is mixed at pickup, record the fix; it closes only after 3 separate days of separate pickups.
8. Report	A printable weekly sheet for the warden or estate office, with a per-area summary. No names, no photos.

No student names. No public rankings. No points, prizes or fines. A person always makes the rating; AI is switched off on the live site.

3. Who uses it

Person	Uses	Needs
Pilot lead (team, warden or estate office)	The main site: checks, weekly plan, cards, reports	The operator access code
Cleaning staff	The worker page on their own phone	Only the worker link
Students and residents	The scorecard page, and My Bin Guide	Only the card link

4. How to use it — the routine

Once

1. Sign in and choose **Your real pilot** (Demo is only for practice).
2. **Pilot setup**: add each bin pair on campus as a location, with its area, with permission.
3. **Pilot setup** → **Collection staff link**: create it and send it to cleaning staff on WhatsApp.

Every day

4. **Record a check** at each location, morning and evening.
5. Cleaning staff report from their phones at pickup.

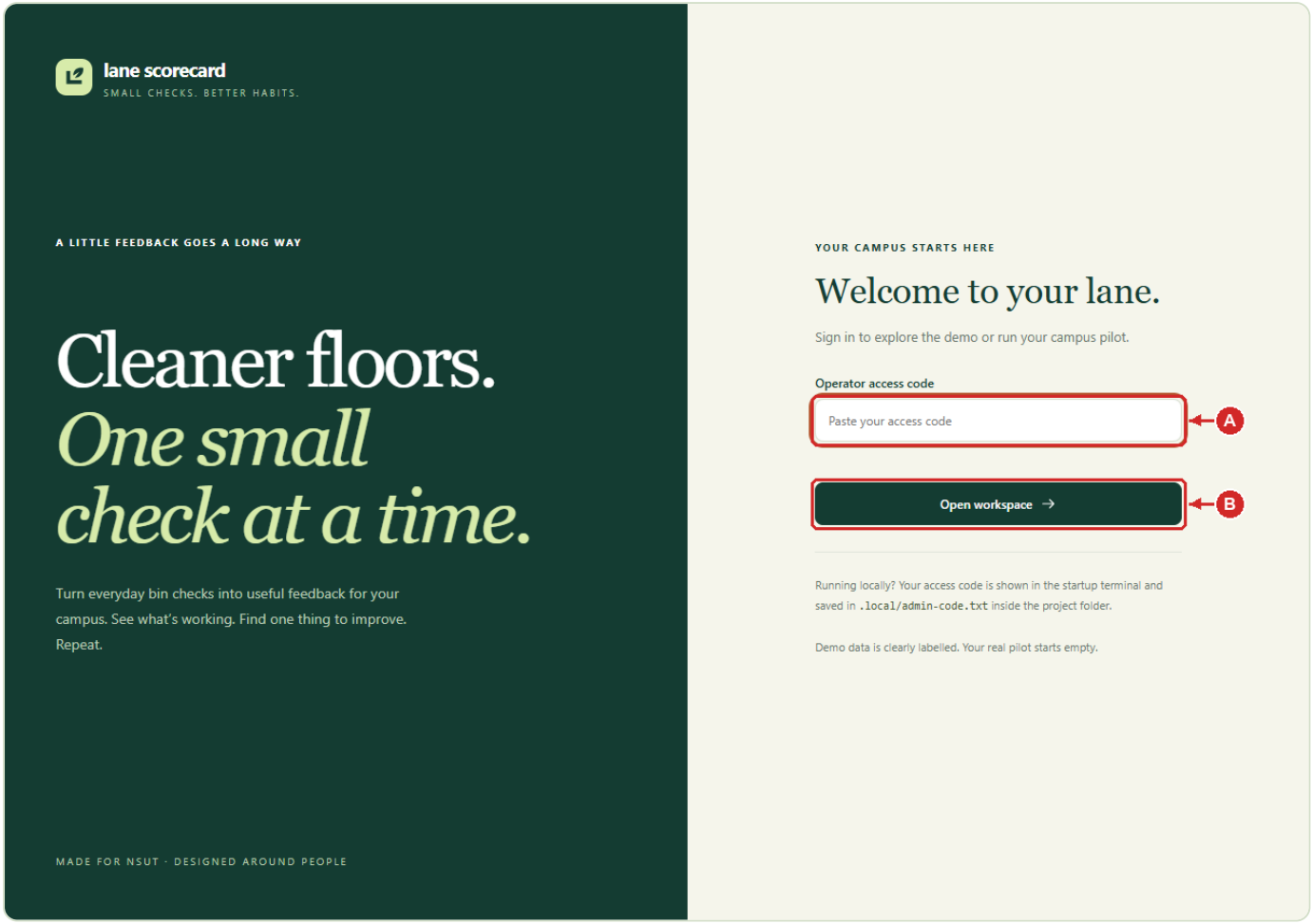
Every week

6. **Weekly plan**: one action per location, with the “vs last week” result.
7. **Scorecards**: make a card per location and share or print it.
8. **Download weekly report** for the warden or campus office.
9. **Observation log**: fix mistakes or download everything as CSV.

> **Rules:** record only what you actually saw · use “Not seen” instead of guessing · no names, faces or room numbers · get permission first · keep test data in Demo, real data in Your real pilot.

Sign in

The warden or volunteer side is protected by the operator access code. Workers and people opening a scorecard never need it.



A Paste the access code here. Get it from the team lead in a private message; never post it in a group.

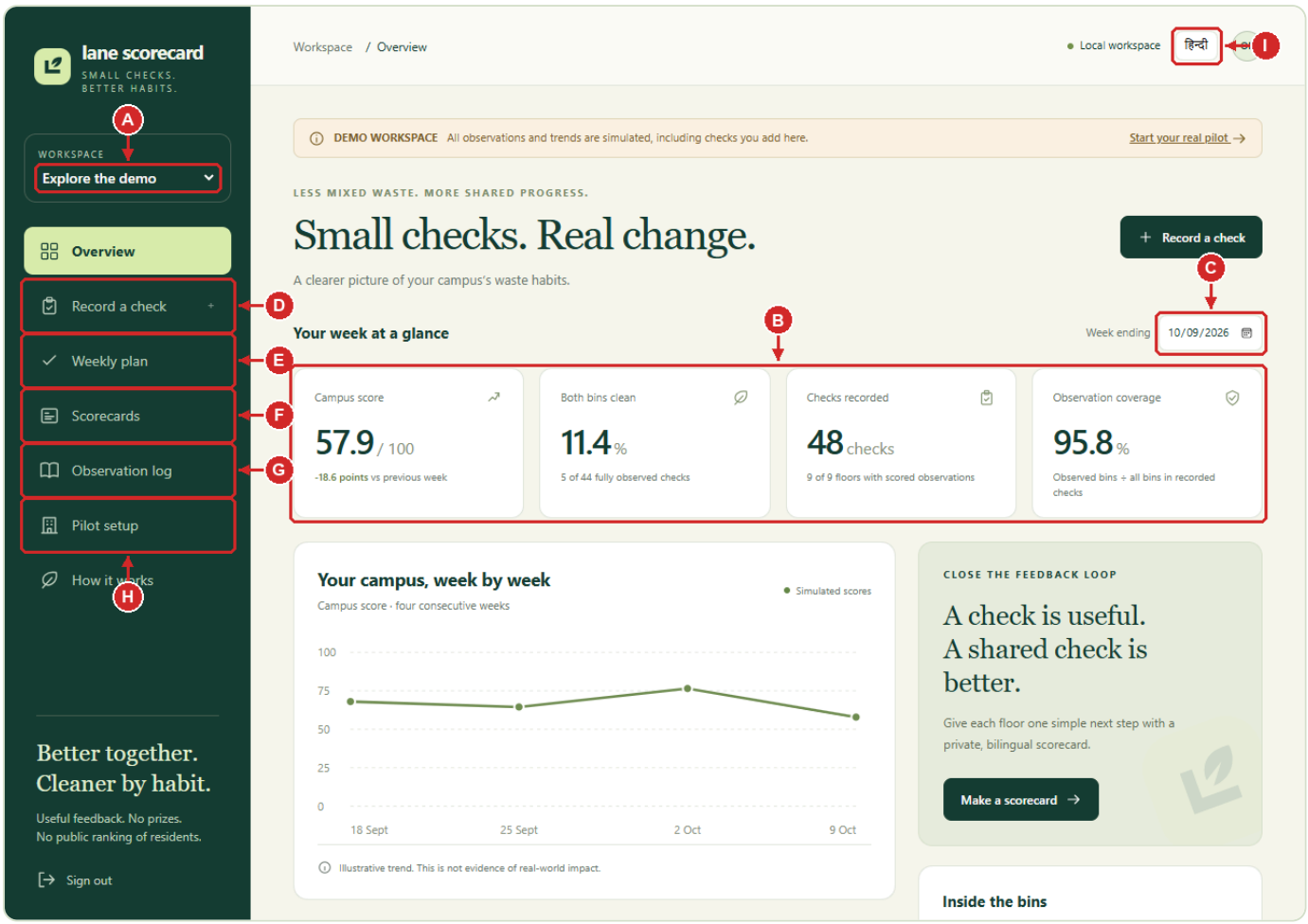
B Sign in. A wrong code shows an error; after 10 wrong tries in 5 minutes the site makes you wait.

Do this

- Open the live site.
- Paste the code and press Sign in.
- Let the browser save the password so you type it only once.

Overview — your week at a glance

The first screen after signing in. This shows the Demo workspace, so every number is simulated and labelled DEMO.



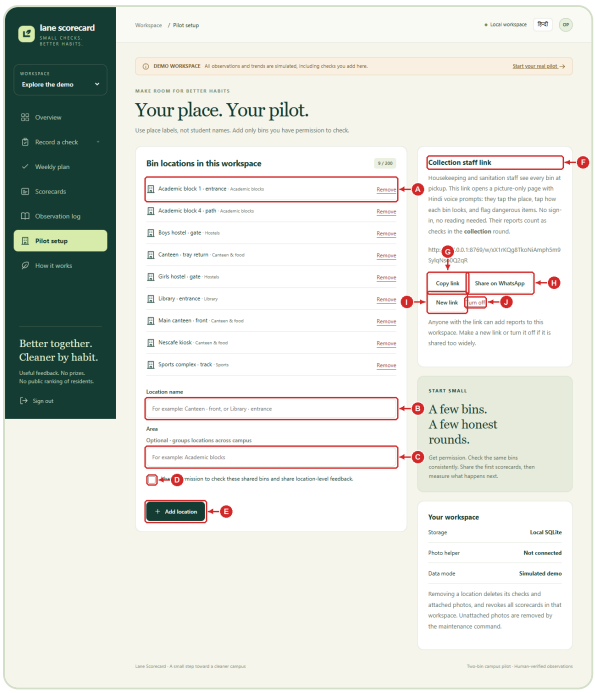
A	Workspace switch. **Demo** = practice data. **Your real pilot** = real checks only. Always check this before recording.
B	The four key numbers: community score out of 100, how often both bins were clean, how many checks, and coverage (how many bins were actually seen).
C	Week ending: pick a date to see any earlier week.
D	**Record a check** — the two-minute form (page 7).
E	**Weekly plan** — one next step per location (page 8).
F	**Scorecards** — make shareable feedback cards (page 11).
G	**Observation log** — every check; correct, delete or download (page 12).
H	**Pilot setup** — add bin locations and the worker link (page 6).
I	हिन्दी / English switch for the whole app.

Do this

- Choose **Your real pilot** before entering real data.
- Use the left menu to move between pages.

Pilot setup — add bin locations and the worker link

Do this once. Each pair of green and blue bins on campus is one “location”, for example “Canteen · front”, grouped into an area.



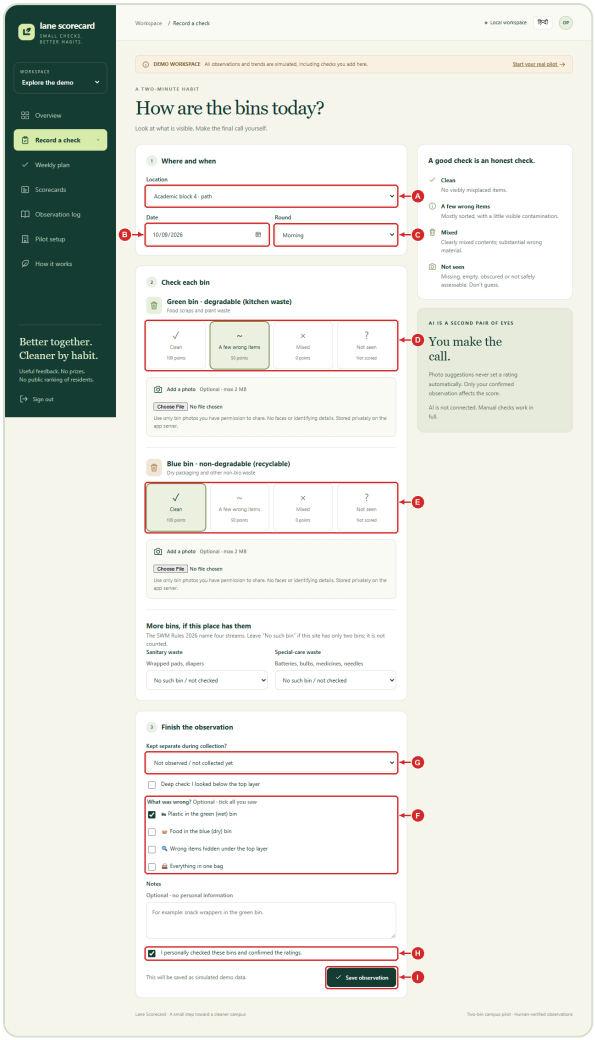
A	Your locations. **Remove** deletes a location and all its checks and cards, so use it with care.
B	Location name: a place label, never a person. Examples: Canteen · front, Library · entrance.
C	**Area** (optional): which part of campus — Academic blocks, Hostels, Canteen & food, Library, Sports… Pick a suggestion or type your own. Locations are grouped by area everywhere.
D	Tick to confirm you have permission to check these bins.
E	**Add location**. Up to 200 per campus.
F	Collection staff link: a secret link for housekeeping or sanitation staff. It opens the picture-only page (page 16). No sign-in.
G	Copy the link.
H	Share it on WhatsApp directly to the staff member.
I	**New link**: the old link stops working at once. Use it if the link was shared too widely.
J	**Turn off**: no one can send reports until you make a new link.

Do this

- Add every bin pair you will check, with its area.
- Create the worker link and send it to the cleaning staff, with their consent.

Record a check — about two minutes

Look at what is visible and rate both bins. Only record what you actually saw.



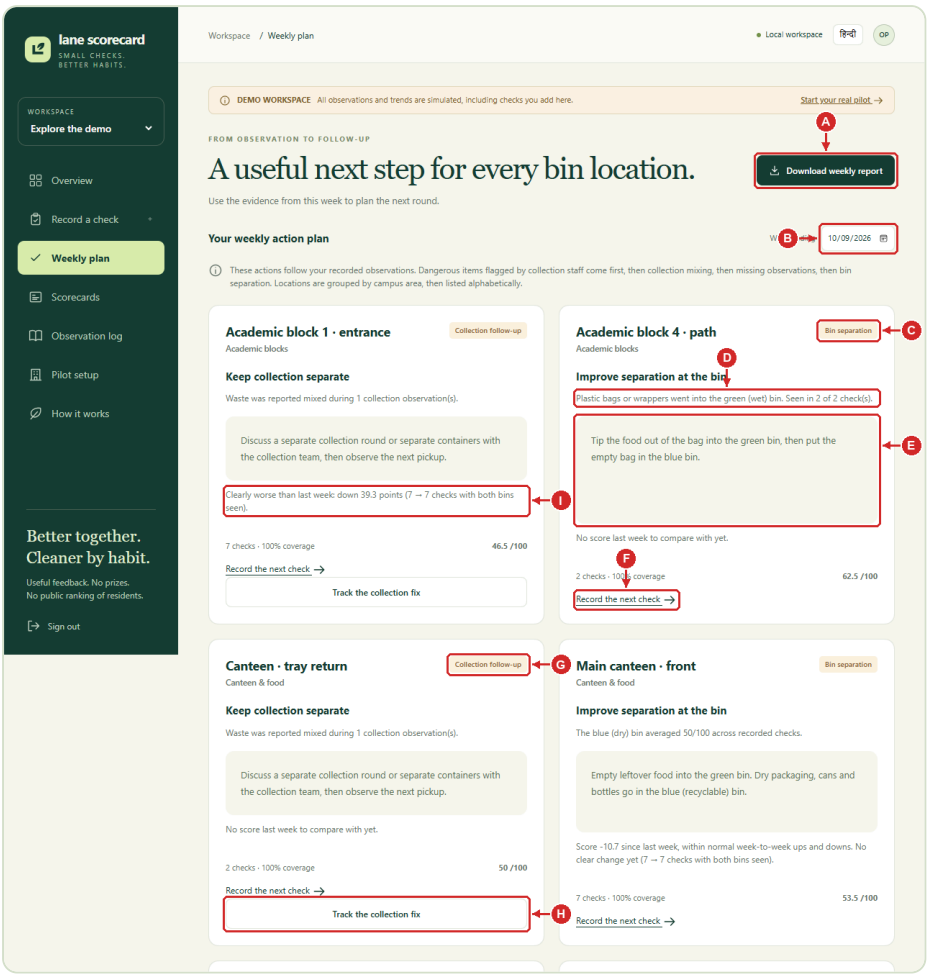
A	Which location.
B	Date (today or earlier, never the future).
C	Round: morning or evening. One check per location, per day, per round.
D	Green bin — degradable (kitchen waste): Clean = 100, A few wrong items = 50, Mixed = 0, Not seen = not counted. Never guess; use Not seen.
E	Blue bin — non-degradable (recyclable), same scale.
G	Kept separate during collection? Only if you saw the pickup. This catches bins being emptied into one cart.
F	What was wrong? Tick what you saw: plastic in the green bin, food in the blue bin, wrong items hidden underneath, or everything in one bag. These power the specific feedback on the card.
H	Confirm you checked the bins yourself. Required.
I	**Save observation**. Scores update straight away.

Do this

- Optional fields (not marked): a bin photo per bin (no faces), extra sanitary / special-care bins if the site has them, “Deep check: I looked below the top layer”, and a short note with no personal details.

Weekly plan — one useful next step for every location

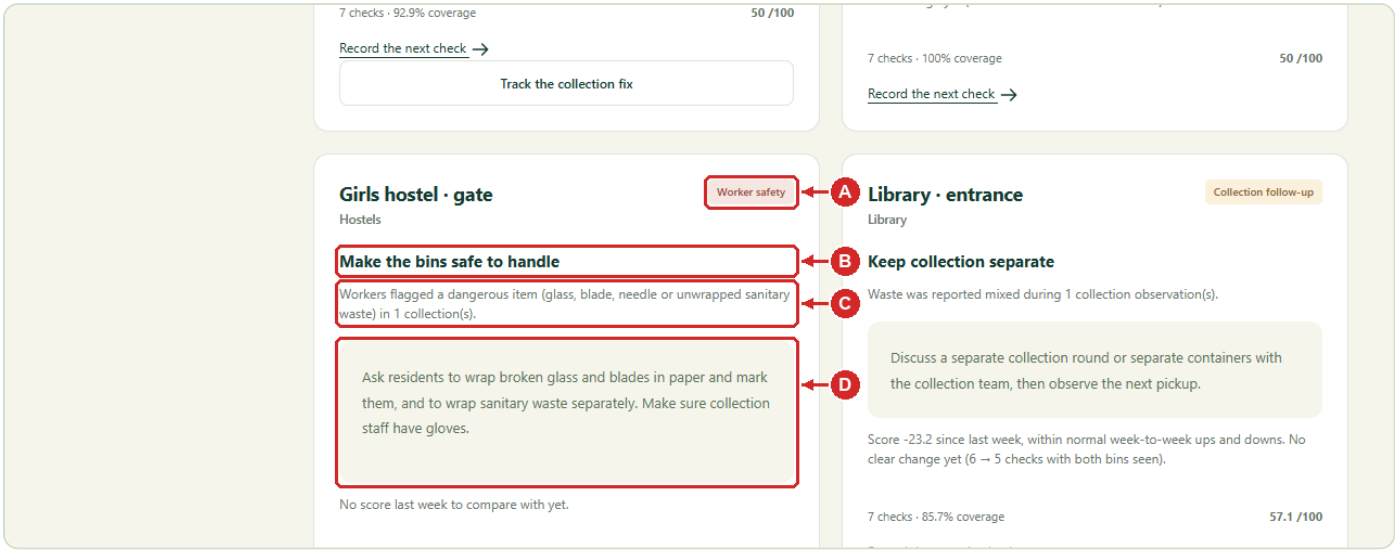
One action per location, with the evidence behind it, grouped by campus area (the grey label under each name). Read it once a week, do the action, keep checking: next week shows whether it worked.



A	**Download weekly report** for the warden or campus office (page 13).
B	Choose the week.
C	What kind of action this is. Order of priority: worker safety → collection mixing → missing observations → bin separation.
D	The evidence, in plain words: what went wrong and how often (“Seen in 2 of 2 checks”).
E	The one thing to do next.
F	**Record the next check** for this location.
G	A collection problem: waste was mixed at pickup.
H	**Track the collection fix** — starts a follow-up (page 10).
I	“Vs last week”: compares only checks where both bins were seen, and calls a change clear only if it is at least 5 points and bigger than normal week-to-week ups and downs. Here a 39-point drop counts as “clearly worse”; smaller drops elsewhere stay “no clear change”.

Worker safety comes first

When cleaning staff flag glass, blades, needles or unwrapped sanitary waste, that location's action changes.



A Marked **Worker safety**.

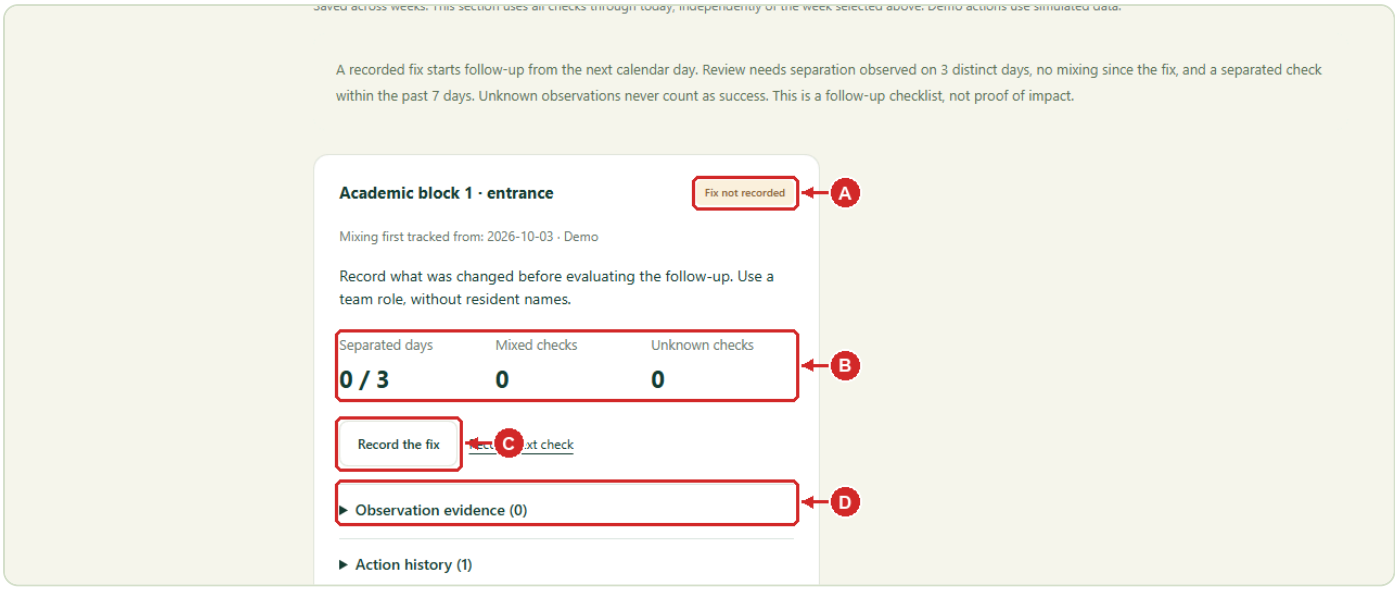
B The action becomes "Make the bins safe to handle", ahead of everything else.

C Evidence: how many collections had a dangerous item.

D What to ask residents, and a reminder that staff need gloves.

Collection follow-up — has the pickup problem really been fixed?

After “Track the collection fix”, the case appears at the bottom of the Weekly plan.



A Status. It starts as **Fix not recorded**.

B Evidence counts: days collection stayed separate (needs 3 different days), mixed checks, and unknown checks.

C **Record the fix**: what was changed, by which role (no names).

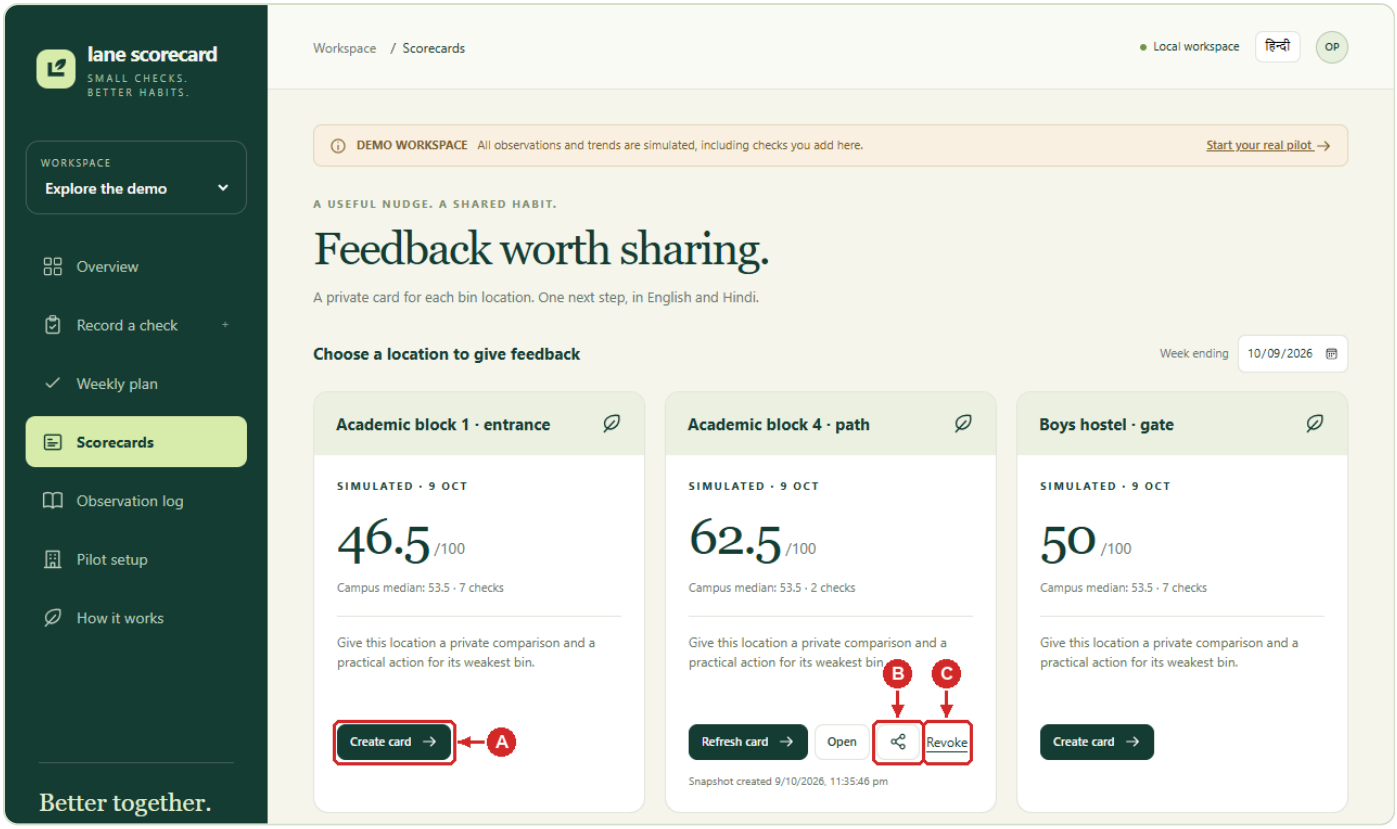
D The checks used as evidence.

Do this

- Record the fix.
- Keep recording whether collection kept waste separate.
- It becomes “Ready for operator review” only after separation on 3 different days and no mixing since the fix. Unknown never counts as success.

Scorecards — feedback people will actually read

A private card for each location, in English and Hindi.



A **Create card** for this week.

B Share or copy the card's private link.

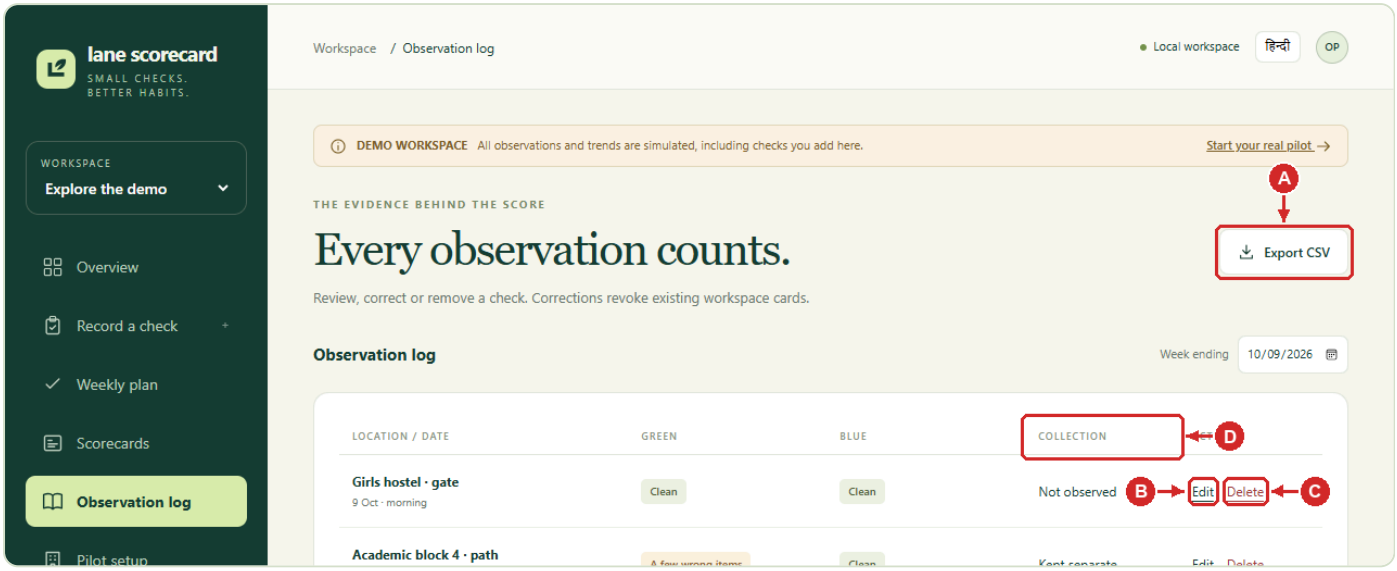
C **Revoke**: the link stops working. Cards are also revoked automatically when data is corrected, so old numbers are never shared.

Do this

- Not marked: **Refresh card** (make a new snapshot) and **Open** (view the card).

Observation log — the evidence behind every score

Every check, newest first.



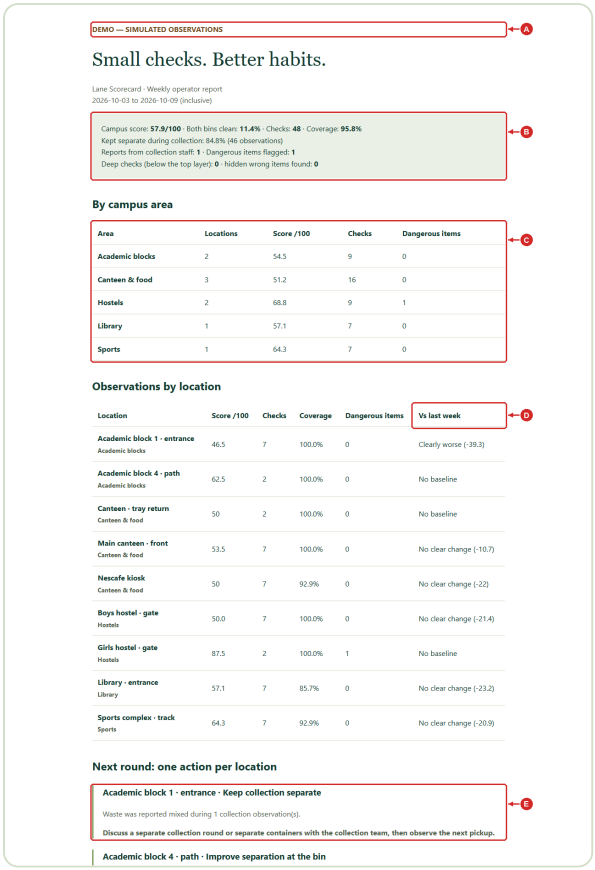
- **A** **Export CSV**: opens in Excel. Includes the hazard, reporter (operator or worker) and issue tags.
- **B** **Edit** a wrong check. Location, date and round stay fixed.
- **C** **Delete** a check.
- **D** Collection column: kept separate, mixed, or not observed.

Do this

- Fix mistakes here, not by recording a second check.

The weekly report — a sheet for the warden

Downloaded from the Weekly plan. Open it in a browser and print or save as PDF.



DEMO — SIMULATED OBSERVATIONS

Small checks. Better habits.

Lane Scorecard · Weekly operator report
2026-10-03 to 2026-10-09 (inclusive)

Campus score: **57.9/100** · Both bins clean: **11.4%** · Checks: **48** · Coverage: **95.8%**
Kept separate during collection: 84.8% (46 observations)
Reports from collection staff: **1** · Dangerous items flagged: **1**
Deep checks (below the top layer): **0** · hidden wrong items found: **0**

By campus area

Area	Locations	Score /100	Checks	Dangerous items
Academic blocks	2	54.5	9	0
Canteen & food	3	51.2	16	0
Hostels	2	68.8	9	1
Library	1	57.1	7	0
Sports	1	64.3	7	0

Observations by location

Location	Score /100	Checks	Coverage	Dangerous items	Vs last week
Academic block 1 · entrance — Academic blocks	46.5	7	100.0%	0	Clearly worse (-39.3)
Academic block 4 · path — Academic blocks	62.5	2	100.0%	0	No baseline
Canteen · tray return — Canteen & food	50	2	100.0%	0	No baseline
Main canteen · front — Canteen & food	53.5	7	100.0%	0	No clear change (-10.7)
Nescafe kiosk — Canteen & food	50	7	92.9%	0	No clear change (-22)
Boys hostel · gate — Hostels	50.0	7	100.0%	0	No clear change (-21.4)
Girls hostel · gate — Hostels	87.5	2	100.0%	1	No baseline
Library · entrance — Library	57.1	7	85.7%	0	No clear change (-23.2)
Sports complex · track — Sports	64.3	7	92.9%	0	No clear change (-20.9)

Next round: one action per location

Academic block 1 · entrance · Keep collection separate

Waste was reported mixed during 1 collection observation(s).

Discuss a separate collection round or separate containers with the collection team, then observe the next pickup.

Academic block 4 · path · Improve separation at the bin

A Says clearly whether the data is DEMO or REAL PILOT.

B Summary: community score, both-bins-clean rate, checks, coverage, collection separation, staff reports, dangerous items, deep checks.

C **By campus area**: locations, average score, checks and dangerous items for each area — see at a glance which part of campus needs attention.

D "Vs last week": clearly better, clearly worse, no clear change, or no baseline.

E One action per location, with its evidence.

Do this

- No notes, photos, names or links are in the report. A downloaded copy does not update after corrections.

The scorecard on a phone — what residents see

Opened from the private link. No sign-in needed.



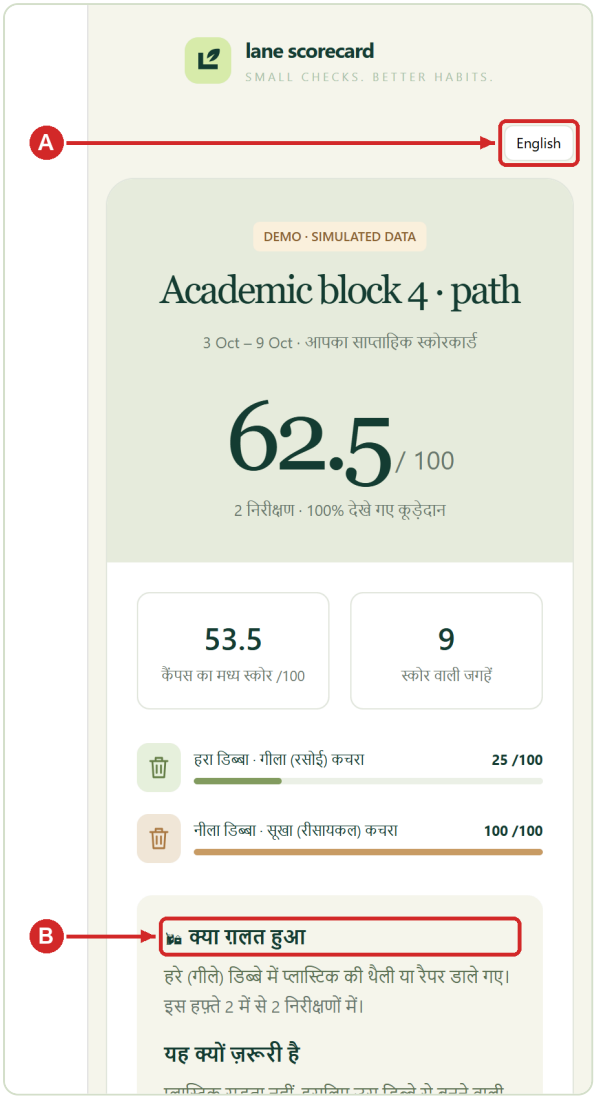
A	This week's score out of 100, with number of checks and coverage.
B	How the community did (median score) and how many locations were compared.
C	Each bin's score.
D	**What went wrong** — the most common mistake, and how often.
E	**Why it matters**.
F	**Other locations** — how many of the others avoided it (shown only with 2+ others).
G	**One thing to try**.
H	हिन्दी / English.
I	Copy private link · "Which bin? Check a receipt" (My Bin Guide) · WhatsApp · Print.

Do this

- Share the link with the people who use that bin, or print it for the notice board.

The same card in Hindi

Every part of the card is available in Hindi.



A Switch back to English.

B "क्या ग़लत हुआ" — What went wrong, in Hindi.

The worker page — for cleaning staff, no reading needed

Opened from the collection staff link on the worker's own phone. Every tap is spoken aloud in Hindi.



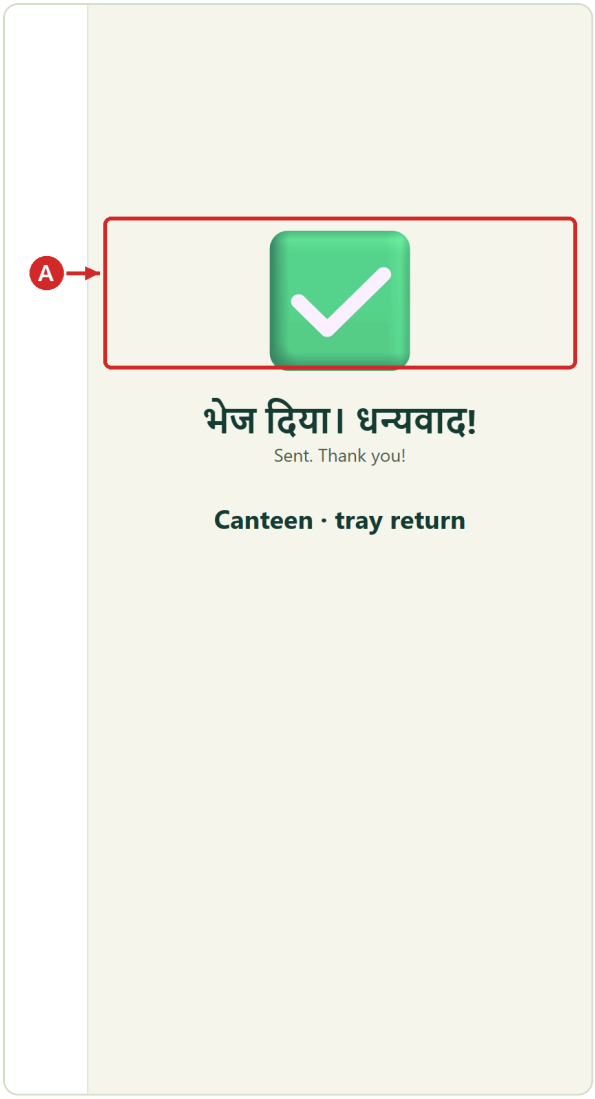
A	🔊 Listen: reads the instructions aloud.
B	Step 1 — tap the place, listed under its campus area. Its name is spoken.
C	Step 2 — the green (wet) bin.
D	Tap a face: 😊 clean, 😐 a few wrong, 😟 mixed.
E	Step 3 — the same for the blue (dry) bin.
F	Step 4 (optional) — what was wrong, as pictures.
G	"I looked below the top layer" — tap if they checked underneath.
H	Step 5 — red danger button: glass, blade, needle or unwrapped sanitary waste.
I	✓ भेजें (Send). Grey until a place and both bin faces are chosen.

Do this

- Saved as today's "collection" check for that place. A second report the same day replaces the first (fixing a mistake).
- The worker sees only place names — no scores, notes or other data.

Report sent

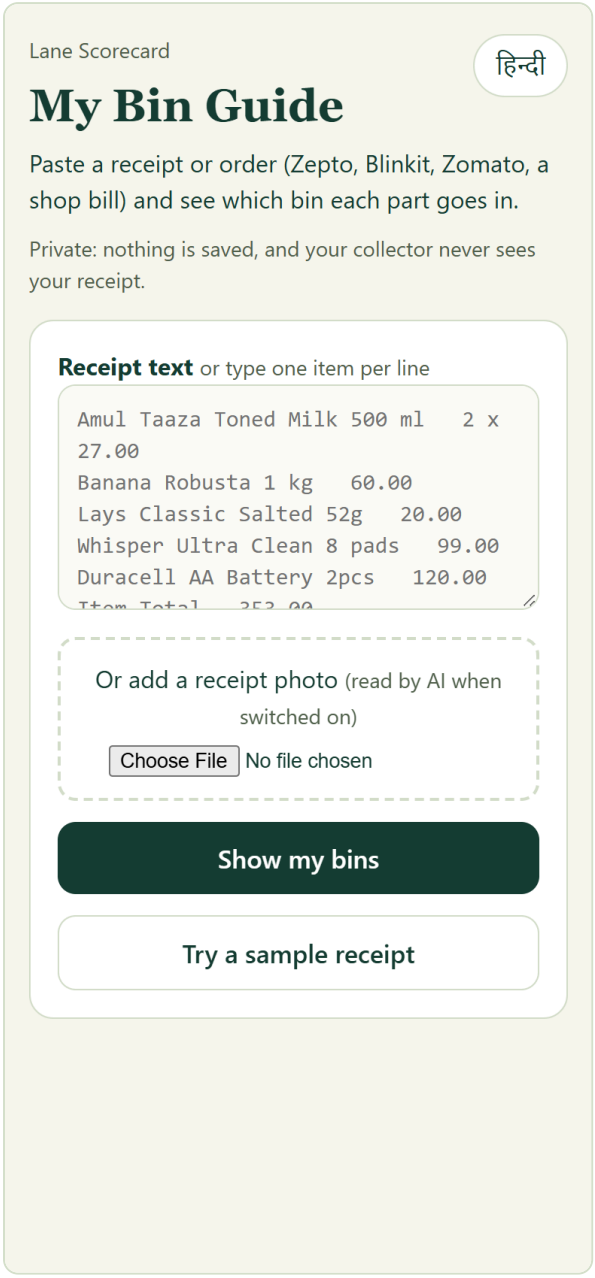
A big tick and “धन्यवाद” (thank you), spoken aloud. The page then resets for the next place.



A Sent successfully.

My Bin Guide — which bin does each item go in?

For students: paste a receipt or order list (Zepto, Blinkit, Zomato, a shop bill) and see which bin each part goes in. Nothing is saved.



Do this

- Reading receipt photos uses AWS AI, which is **switched off** on the live site, so the page says AI is not available for photos. Typed or pasted text still works with the reviewed rules.

What it is not

- It does **not** prove that waste went down. It shows what is happening and whether it changes.
- It does **not** use AI to decide ratings. A person always decides.
- It does **not** fine, rank or shame anyone.

Where it runs

On AWS in us-east-1: API Gateway gives the web address, Lambda runs the app, DynamoDB stores the data, S3 holds photos privately, EventBridge makes the weekly cards and CloudWatch keeps the logs.

If something looks wrong

“That access code is not correct”	Check for spaces when pasting; ask the team lead for the current code.
The worker link says it is not working	A new link was made or the link was turned off. Ask the warden for the current link.
No sound on the worker page	The phone has no Hindi voice. The page still works by pictures.
A check was recorded in Demo by mistake	Demo is only practice; record it again in Your real pilot.
A number looks wrong	Open the Observation log, check the entries, and Edit any mistake.